



# Modeling Anti-NMDAR Encephalitis Using Petri Nets

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# Outline

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**Bio-modeling** *Petri net definitions, decisions, and the two coupled modules*

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**Analysis and Results** *Validation of baseline vs. perturbation dynamics*

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**Conclusions** *Findings, limitations, and future work*

## Section 1

# Biological Background

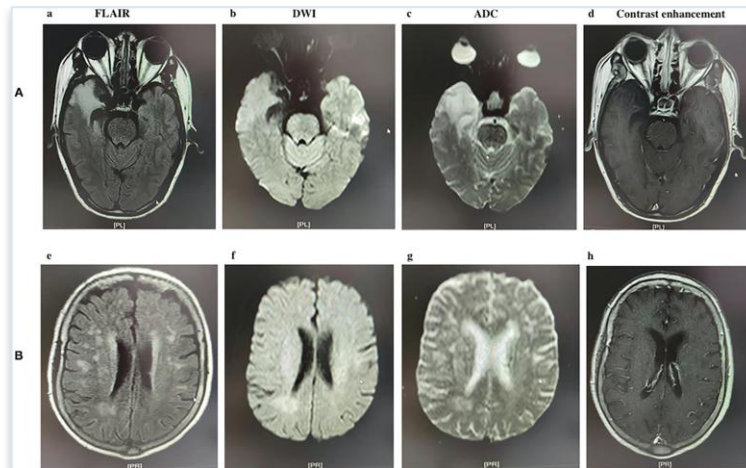
Encephalitis, the NMDA receptor, and the autoimmune mechanism

02



# Anti-NMDAR Encephalitis

- Autoimmune encephalitis: **inflammation** of the brain caused by patient **IgG antibodies** targeting the **GluN1 subunit** of NMDARs
- Antibodies bind → crosslink → internalize receptors → lysosomal degradation
- **Result:** fewer surface NMDARs → less  $\text{Ca}^{2+}$  enters the neuron
- **Symptoms:** psychosis, memory loss, seizures, movement disorders



Brain MRI in anti-NMDAR encephalitis: temporal-lobe and white-matter T2 / FLAIR lesions. Source: Front. Neurol. 2022, Fig. 1 (CC BY).

# What are we modeling?

- We model the downstream consequences of NMDAR loss using a **discrete stochastic Petri net** (Gillespie SSA), comparing:

- **Baseline:**

healthy NMDAR signalling → normal  $\text{Ca}^{2+}$  influx → CaMKII activation, LTP, neuronal survival

- **Perturbation:**

antibody-driven NMDAR internalisation → reduced  $\text{Ca}^{2+}$  → three divergent stress branches:

- **ER stress** → CHOP induction → pro-apoptotic signalling
  - **Mitochondrial  $\text{Ca}^{2+}$  imbalance** → ROS accumulation → oxidative stress
  - **Convergence on CASP3** → apoptosis → neurodegeneration
- **Research question:** how does antibody-mediated NMDAR depletion alter the dynamic balance between calcium-dependent survival and apoptotic signalling?

## Section 2

# Bio-modeling

Petri net definitions, modelling decisions, and the two modules

03



# A discrete stochastic Petri net (SPN)

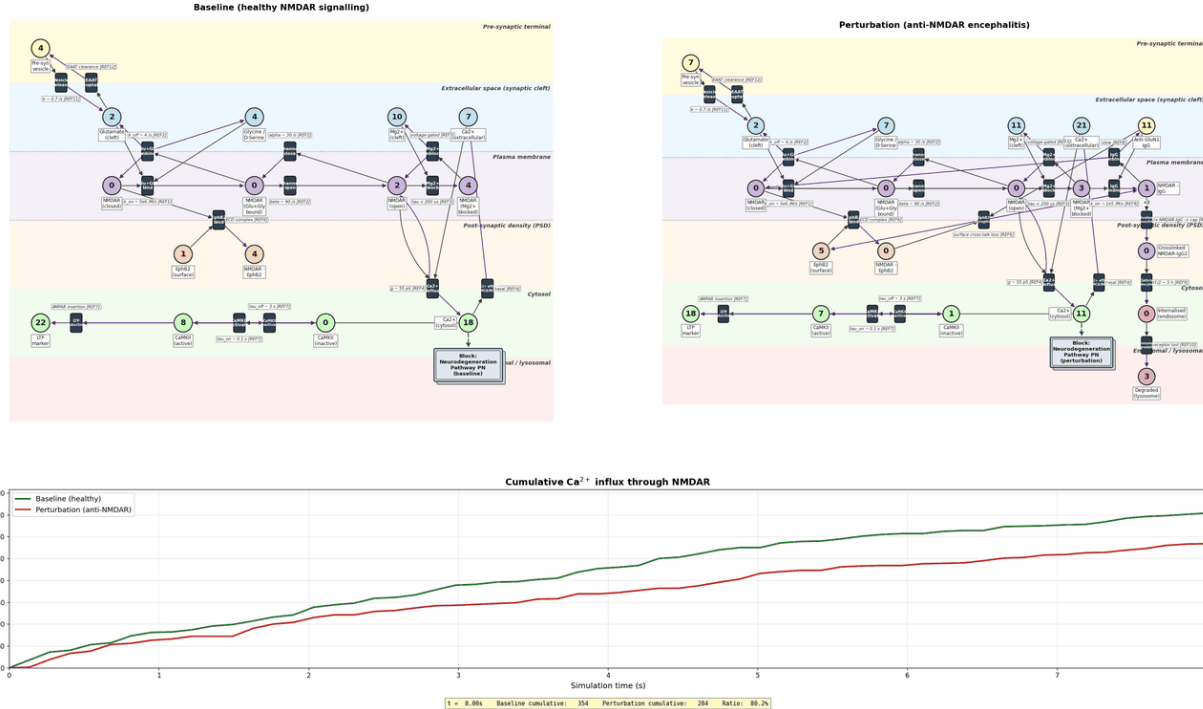
- We model NMDAR signalling at a glutamatergic synapse as a discrete stochastic Petri net.
- **A Petri net is a 5-tuple  $N = (P, T, F, W, M0)$ :**
  - $P$  places - molecular species or states,
  - $T$  transitions - biochemical reactions or events,
  - $F$  flow relation - directed arcs between places and transitions,
  - $W$  arc weights - stoichiometry,
  - $M0$  initial marking - tokens per place.
- Stochastic firing via the Gillespie SSA: each transition fires with a mass-action propensity over its input tokens.
- A transition is enabled when every input place holds at least the arc weight; one firing per SSA step; ordinary arcs only.

## Propensity (mass action)

- $a(t) = k \cdot \text{product over input places of } m(p)! / [(m(p) - w(p))! \cdot w(p)!]$
- $k$  is the rate constant,  $m(p)$  the marking of input place  $p$ ,  $w(p)$  the arc weight.
- Built in a small, fully introspectable Python Petri-net core.
- Exported to Snoopy (.spn / .cpn) and Charlie (APNN / ANDL) for external analysis.

# Two coupled nets sharing cytosolic Ca<sup>2+</sup>

Anti-NMDAR Encephalitis: Petri Net Models and Cumulative Ca<sup>2+</sup> Influx



Side-by-side overview: baseline (left) and perturbation (right) NMDAR nets with the cumulative Ca<sup>2+</sup> curve.



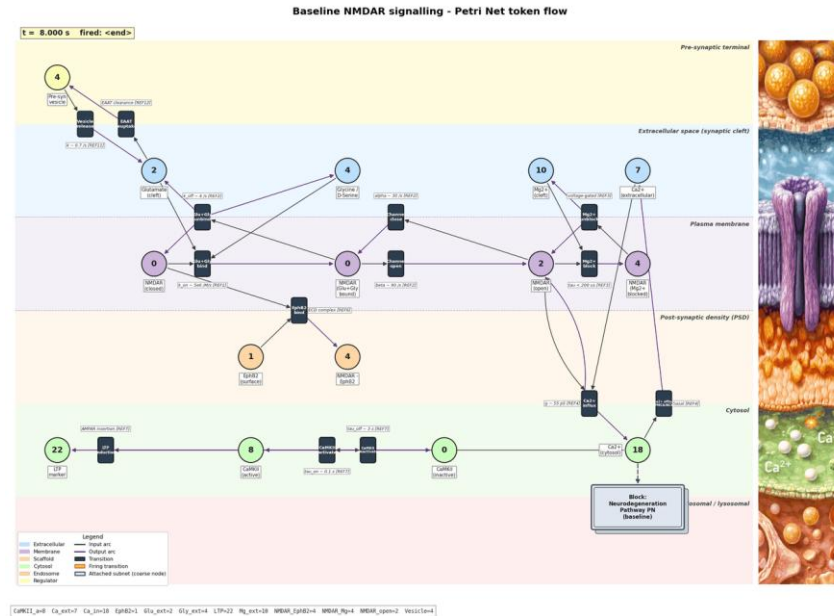
# Two coupled nets sharing cytosolic Ca<sup>2+</sup>

- The model is split into two coupled SPNs joined by a shared cytosolic Ca<sup>2+</sup> place - the interface between synaptic signalling and downstream neurodegeneration.
- Each module has a baseline and a perturbation version; all are simulated for 100 Gillespie SSA runs.
- Perturbation adds antibody dynamics ( $k_{\text{on}} \sim 1e5$  /M/s), bivalent crosslinking, and lysosomal degradation, progressively losing surface NMDARs.
- Faithful comparison: identical core initial conditions, so every divergence comes from the disease-specific machinery alone.

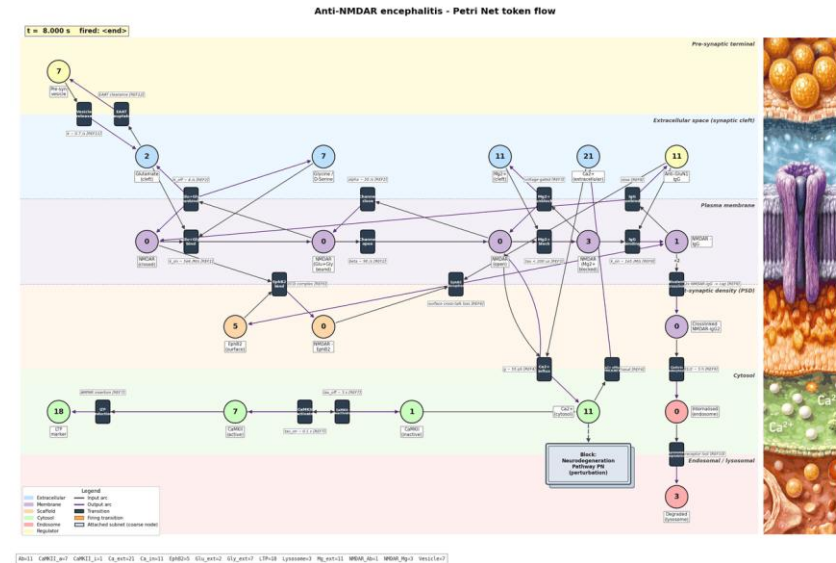
## What the perturbation adds

- NMDAR + Anti-GluN1 IgG -> NMDAR-IgG (T\_Ab\_bind)
- 2x NMDAR-IgG -> crosslinked cap (T\_crosslink)
- NMDAR-EphB2 + IgG -> EphB2 disruption (Mikasova 2012)
- crosslinked cap -> endosome (T\_endocytosis)
- endosome -> lysosome (receptor pool lost)
- Net effect: surface NMDAR pool drains -> Ca<sup>2+</sup> influx and the CaMKII / LTP cascade collapse.

# The glutamatergic synapse, healthy and perturbed

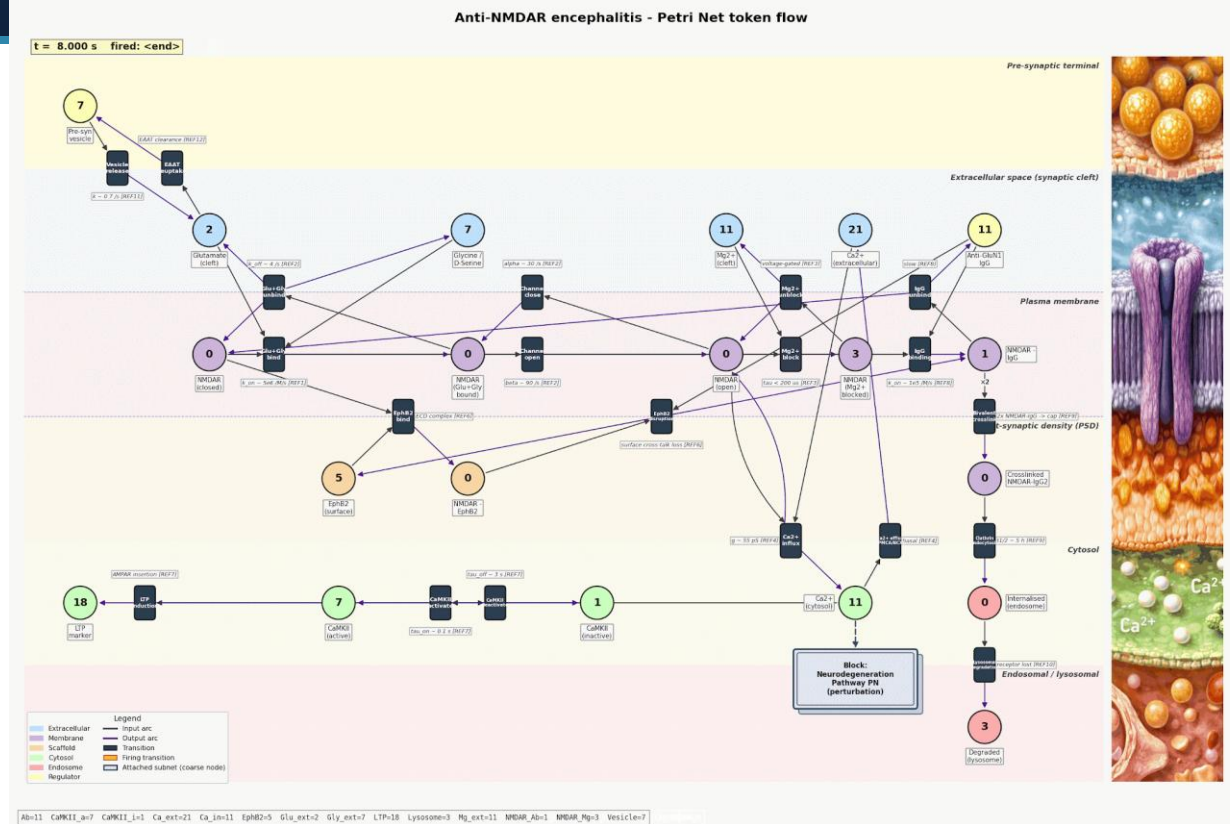


Baseline: healthy NMDAR signalling Petri net (animated).



*Perturbation: anti-NMDAR net with IgG binding, crosslinking, endocytosis, and lysosomal degradation (animated).*

# The glutamatergic synapse perturbed



*Perturbation: anti-NMDAR net with IgG binding, crosslinking, endocytosis, and lysosomal degradation (animated).*

# The glutamatergic synapse, healthy and perturbed

- Models presynaptic vesicle release, glutamate + glycine binding, voltage-dependent  $Mg^{2+}$  block / unblock, channel opening,  $Ca^{2+}$  influx and extrusion (PMCA / NCX), and CaMKII / LTP activation.
- The perturbation variant adds antibody-mediated internalisation: tokens drain from NMDAR (closed) into Degraded (lysosome), lowering the  $Ca^{2+}$  influx events that feed the downstream net.

# Places and initial markings

| Place                   | Baseline | Perturbation |
|-------------------------|----------|--------------|
| Pre-synaptic vesicle    | 8        | 8            |
| Glutamate (cleft)       | 4        | 4            |
| Glycine / D-Serine      | 10       | 10           |
| Mg2+ (cleft)            | 14       | 14           |
| Ca2+ (extracellular)    | 80       | 80           |
| NMDAR (closed)          | 10       | 10           |
| EphB2 (surface)         | 5        | 5            |
| PSD-95 (free)           | 6        | 6            |
| CaMKII (inactive)       | 8        | 8            |
| Ca2+ (cytosol)          | 2        | 2            |
| Anti-GluN1 IgG          | -        | 18           |
| Internalised (endosome) | -        | 0            |
| Degraded (lysosome)     | -        | 0            |

## Reading the table

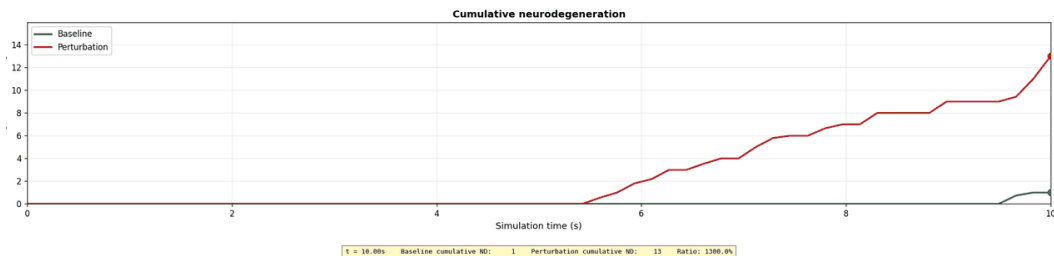
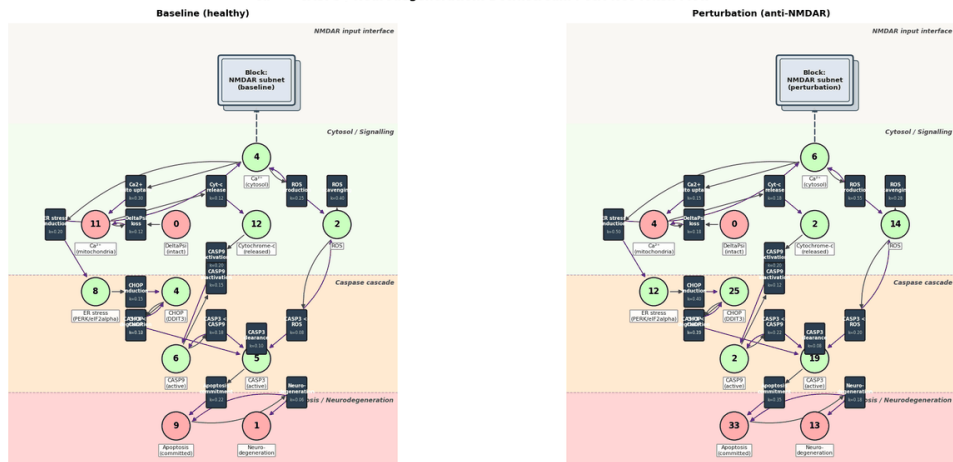
- Both nets share identical core synaptic conditions.
- The perturbation adds three places (shown as '-' in baseline): Anti-GluN1 IgG (18, the patient antibody pool), Internalised (endosome), and Degraded (lysosome).
- Tokens flow NMDAR (closed) -> internalisation -> Degraded, depleting the surface pool and the rate of Ca2+ influx.
- Any downstream divergence is therefore due to antibody-driven internalisation, not different starting conditions.

# From cytosolic $\text{Ca}^{2+}$ to neurodegeneration

- **Extension of the NMDAR model:** cytosolic  $\text{Ca}^{2+}$  place is the interface between the two subnets
- Three parallel stress branches, all converging on CASP3:
  - **Mitochondrial pathway:**  $\text{Ca}^{2+}$  overload  $\rightarrow$  mitochondrial damage  $\rightarrow$  cytochrome c release  $\rightarrow$  CASP9 activation
  - **ER stress pathway:** ER stress  $\rightarrow$  PERK/eIF2 $\alpha$  activation  $\rightarrow$  CHOP induction
  - **ROS pathway:** Excess ROS  $\rightarrow$  oxidative stress  $\rightarrow$  cell damage/death
- **CASP3 activation**  $\rightarrow$  apoptosis  $\rightarrow$  neurodegeneration firing transition
- **Two variants:** baseline ( $\text{Ca}^{2+} = 35$ ) and perturbation ( $\text{Ca}^{2+} = 15$ )

# From cytosolic Ca<sup>2+</sup> to neurodegeneration

Ca<sup>2+</sup> → CASP3 / Neurodegeneration: Downstream Petri Net Token Flow



Downstream Petri nets at  $t = 0$ : baseline (left, Ca<sup>2+</sup> = 35) and perturbation (right, Ca<sup>2+</sup> = 15) (animated).

# Perturbed rate constants and markings

| Parameter                    | Baseline | Perturbation |
|------------------------------|----------|--------------|
| Initial Ca2+ (cytosol)       | 35       | 15           |
| ER stress induction (k)      | 0.20     | 0.50         |
| Ca2+-mito uptake (k)         | 0.30     | 0.15         |
| CHOP induction (k)           | 0.15     | 0.40         |
| Apoptosis commitment (k)     | 0.22     | 0.35         |
| Neurodegeneration firing (k) | 0.06     | 0.18         |

## Why these changes

- **Lower initial Ca2+** (15 vs. 35) reflects antibody-mediated receptor loss upstream.
- **Higher ER-stress induction** (0.50 vs. 0.20) reflects enhanced cellular stress sensitivity under NMDAR dysfunction.
- **Reduced Ca2+-mito uptake** (0.15 vs. 0.30) follows the lower cytosolic Ca2+.
- All other rate constants are held identical, isolating the effect of altered Ca2+ signalling



## Section 3

# Analysis and Results

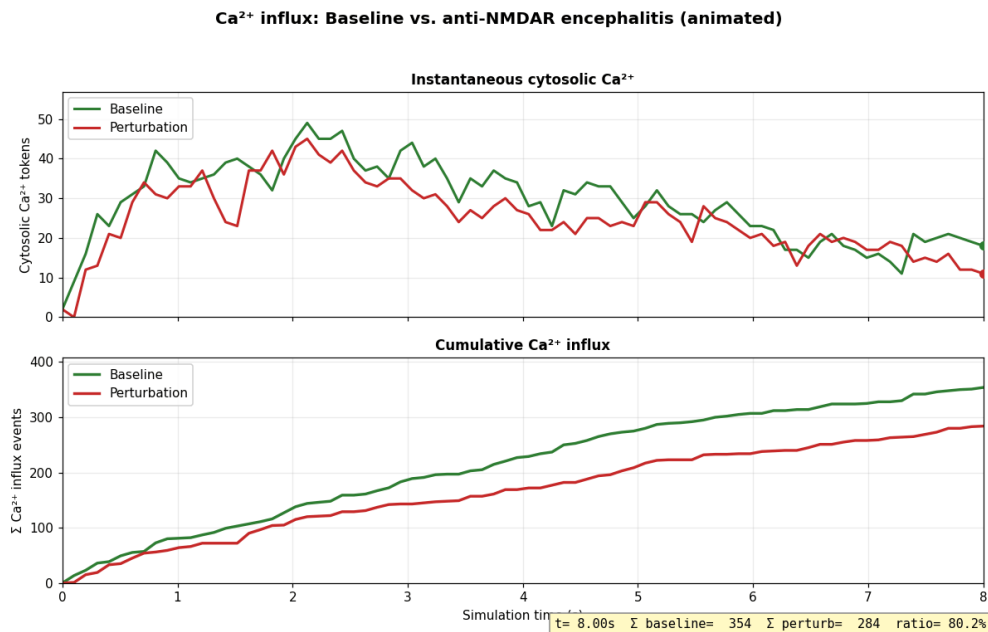
Validating baseline against the anti-NMDAR perturbation

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# Calcium influx: healthy vs. anti-NMDAR

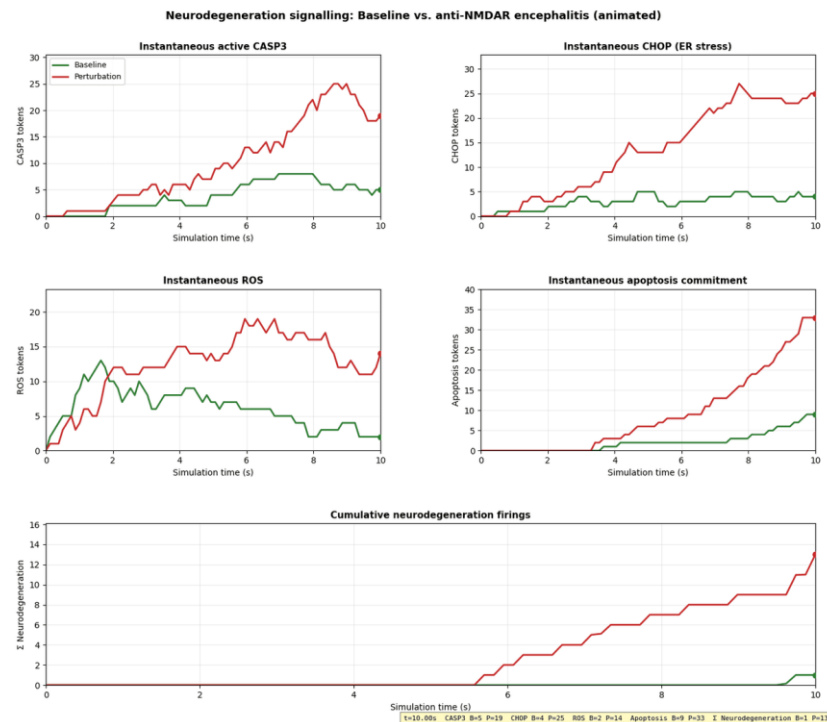
- Healthy signalling (green) sustains higher cytosolic  $\text{Ca}^{2+}$  and greater cumulative influx than the perturbation (red).
- After 8 s, cumulative  $\text{Ca}^{2+}$  influx in the perturbation falls to **80.2% of baseline**,
- This collapse is the input signal handed to the downstream apoptotic module.



Instantaneous cytosolic  $\text{Ca}^{2+}$  (top) and cumulative  $\Sigma \text{Ca}^{2+}$  in firings (bottom); perturbation reaches 80.2% of baseline.

# Downstream trajectories

- A single SSA run over 10 seconds, baseline in green and perturbation in red.
- Every marker is higher under perturbation: active CASP3, CHOP, ROS, apoptosis commitment, and cumulative neurodegeneration.
- The two conditions track together early and start to diverge after about 2 seconds.
- By 10 seconds, neurodegeneration firings reach 13 in the perturbation against 1 in baseline, a thirteen-fold increase.



Single SSA run over 10 s: CASP3, CHOP, ROS, apoptosis tokens, and cumulative neurodegeneration firings.

# Ensemble comparison across 100 runs

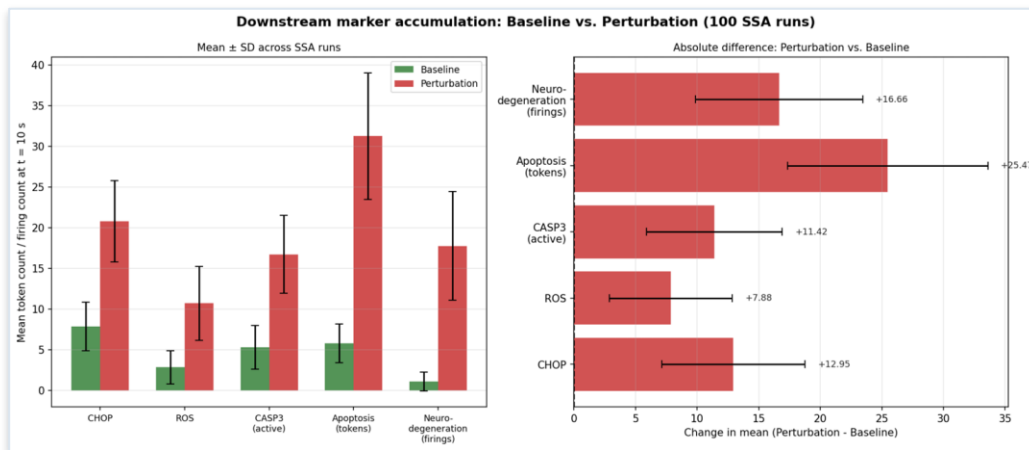
- Marker accumulation at  $t = 10$  s across 100 SSA runs (mean  $\pm$  SD).

- All markers are significantly elevated under perturbation.

- **Downstream effects by rank:**

- Apoptosis
- CHOP
- Neurodegeneration
- CASP3
- ROS

- This supports the hypothesis: NMDAR depletion shifts the balance toward apoptosis and to neurodegeneration through activation of CHOP.



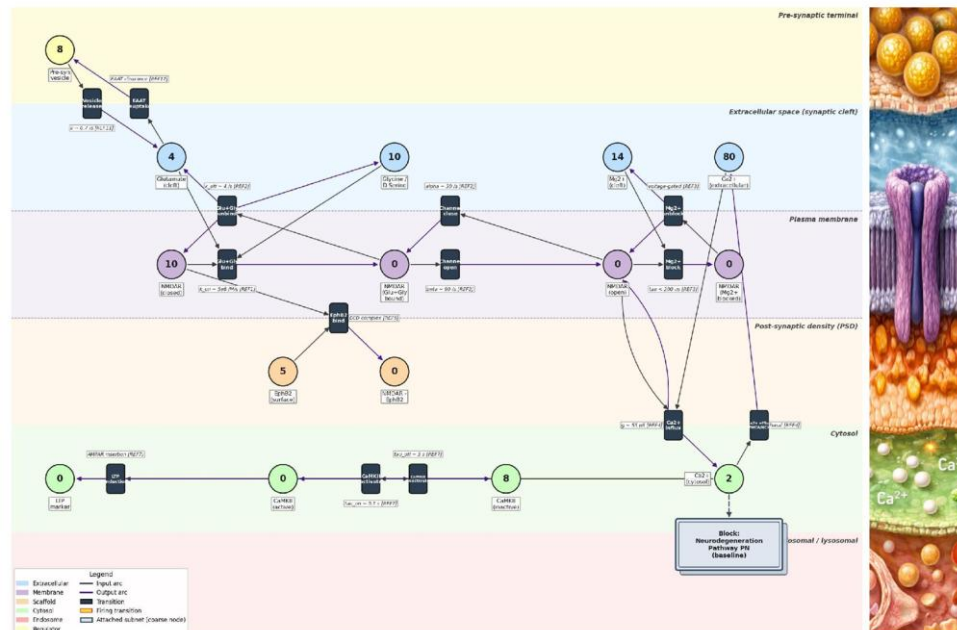
Mean  $\pm$  SD marker counts (left) and the perturbation - baseline difference (right) over 100 runs.

# NMDAR synaptic module: baseline

BASELINE

## Baseline: Healthy NMDAR signalling

Glutamatergic synapse around a single NMDAR pool. Co-agonists (glycine / D-serine), voltage-dependent Mg<sup>2+</sup> block, Ca<sup>2+</sup> influx, PSD-95 scaffolding, EphB2 stabilisation and downstream CaMKII  $\rightarrow$  LTP cascade.



Healthy NMDAR signalling, 15 places and 14 transitions

## Structural properties

- ✓ ORD ordinary
- ✗ CSV conservative
- ✓ FT0
- ✓ TF0
- ✗ PUR pure
- ✓ SB bounded
- ✗ FP0
- ✗ PF0

## P-invariants: six conserved pools

Receptor 10, CaMKII 8, Mg 14, glycine 10, glutamate 12, EphB2 5

## T-invariants: five reversible cycles

release/reuptake, bind/unbind, open/close, Mg block/unblock, Ca in/out

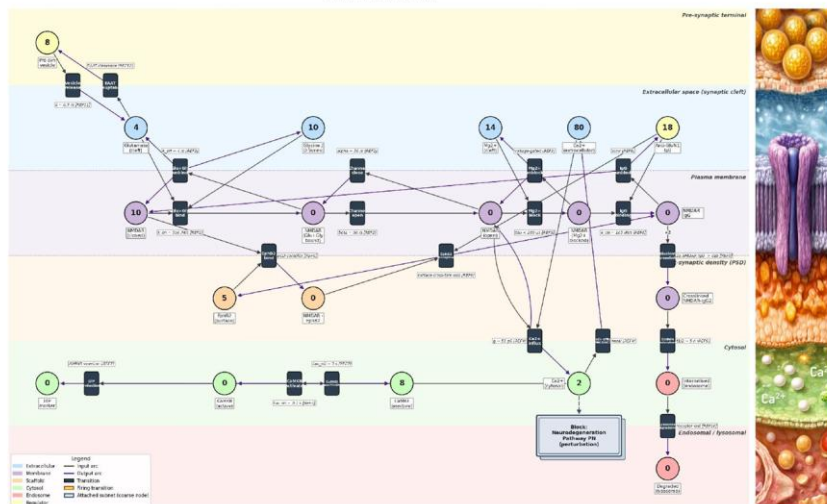
**Calcium and LTP sit in no P-invariant.** Calcium is spent as a signal, so it stays bounded but is never conserved. The net is structurally bounded, so its state space is finite, just far too large to write out.

# NMDAR synaptic module: perturbation

PERTURBATION

## Perturbation: anti-NMDAR encephalitis

Patient IgG against the GRIN1 amino-terminal domain binds surface NMDARs, is bivalently crosslinked, displaces the GluR2-NMDAR interaction (Mokkapa 2012), and drives clathrin-mediated endocytosis → lysosomal degradation (Hughes 2010; Moscati 2016). Net effect: progressive loss of synaptic NMDARs and a net  $\text{Ca}^{2+}$  influx.



anti-NMDAR encephalitis: adds IgG binding, crosslinking and internalisation. 20 places and 20 transitions

## Structural properties

- ✗ ORD ordinary
- ✗ CSV conservative
- ✓ FTO
- ✓ TFO
- ✗ PUR pure
- ✓ SB bounded
- ✓ FPO
- ✗ PFO

**ORD is false** because crosslinking consumes two antibody-bound receptors to make one complex.

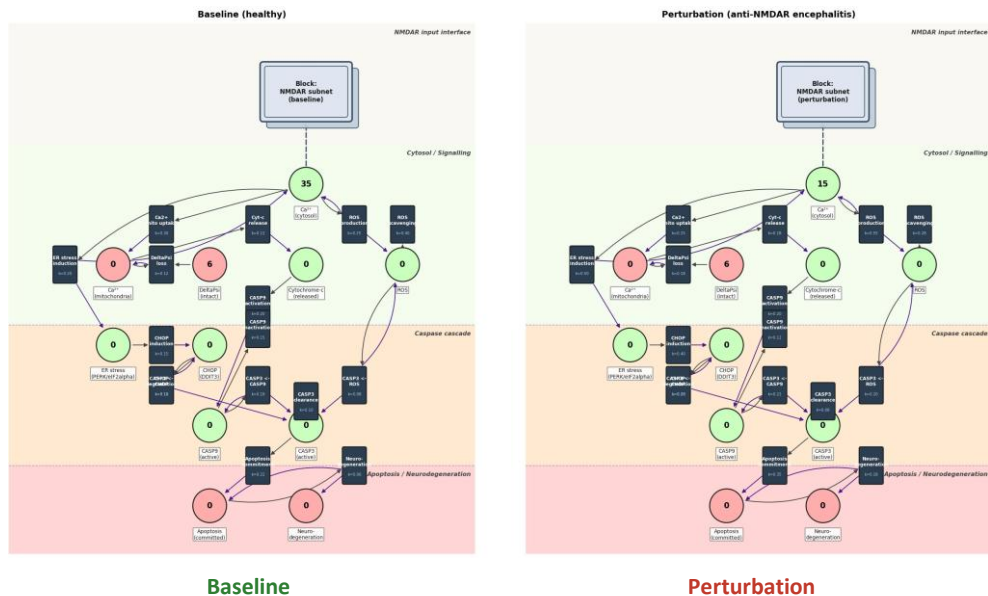
## P-invariants: seven, two now weighted

Five unchanged (CaMKII, Mg, glycine, glutamate, EphB2). Antibody = 18 and Receptor = 10 now carry factors of two on the crosslinked, endosome and lysosome places.

## T-invariants: seven

**Receptor total is still 10.** The receptors are not destroyed, they drain into the lysosomal sink. The one-way steps (crosslinking, endocytosis, lysosomal degradation, CaMKII, LTP) appear in no T-invariant.

# Downstream apoptotic module: baseline and perturbation



$Ca^{2+}$  to CASP3 cascade, 11 places and 16 transitions

## Structural properties

- ✓ ORD ordinary
- ✗ PUR pure
- ✗ CSV conservative
- ✗ SB bounded
- ✓ FT0
- ✗ TF0
- ✗ FP0
- ✗ PF0

**No P-invariants: nothing is conserved.** This is an open cascade from a calcium source to apoptosis and neurodegeneration sinks. Not structurally bounded, so the state space is infinite for.

**Baseline and perturbation are structurally identical.** They share five reversible stress loops and the same one-way mitochondrial and commitment steps. They differ only in rate constants and initial marking of calcium intake, so the divergence we measure is from the initial marking and rates, not structural.

# Limitations and future work

## Limitations

Mass-action rates are scaled for an 8-10 s simulation window, not absolute physiological constants.

Sparse quantitative data exist for many transition rates.

The net contains many truncated pathways, limiting biological realism.

State space of NMDAR pathway is computationally expensive and the neurodegenerative pathway is infinite, therefore traps and sinkholes could arise which we cannot without extensive computational resources.

## Future work

Estimate rates on the critical subgraph and move to a continuous (ODE) net where data allow.

Add inhibitor / read arcs and coloured tokens for richer logic.

Titre-dependent antibody dosing to model disease severity.

Couple the model to treatment scenarios (antibody removal / immunotherapy).



## Section 4

# Conclusions

Findings, limitations, and where this could go next

05



# Key take-home messages

1

## Reduced calcium uptake is the trigger

Antibody internalisation lowers calcium influx through the NMDAR, and that drop is the single signal that drives the whole downstream response.

2

## The damage is large and reproducible

Every stress and apoptotic marker rises under perturbation. Neurodegeneration goes up about thirteen-fold, and the effect holds across 100 runs.

3

## The NMDAR module changes structurally

Going from baseline to perturbation adds the antibody pathway. The net becomes non-ordinary (crosslinking is a weight-two step) and gains a conserved antibody pool, while receptors stay conserved but drain into the lysosomal sink.

4

## The model has limits

The model shows differential of baseline vs. perturbation. Biological relevance is hard to proof due to many truncations and alterations made to the pathway. Extension is needed to provide more biological realism

# Key references

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